

27 Supersymmetric Gauge Theories

27.4 Renormalizable Gauge Theories with Chiral Superfields

We will now put together the pieces assembled in the previous three sections, to construct the most general renormalizable action for chiral superfields interacting with general gauge fields. Adding the terms (27.1.27), (27.2.7), and (27.3.1) and the superpotential terms in Eq. (26.4.5) gives the Lagrangian density

$$\begin{aligned}
\mathcal{L} &= \frac{1}{2} \left[\Phi^\dagger \exp \left(-2 \sum_A t_A V_A \right) \Phi \right]_D - \frac{1}{2} \operatorname{Re} \sum_A (W_{AL}^T \epsilon W_{AL})_{\mathcal{F}} \\
&\quad - \frac{g^2 \theta}{16\pi^2} \sum_A \operatorname{Im} (W_{AL}^T \epsilon W_{AL})_{\mathcal{F}} - \sum_A \xi_A [V_A]_D + 2 \operatorname{Re}[f]_{\mathcal{F}} \\
&= - \sum_n (D_\mu \phi)_n^* (D^\mu \phi)_n - \frac{1}{2} \sum_n (\bar{\psi}_n \gamma^\mu (D_\mu \psi_L)_n) + \sum_n \mathcal{F}_n^* \mathcal{F}_n \\
&\quad - \operatorname{Re} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} (\psi_{nL}^T \epsilon \psi_{mL}) + 2 \operatorname{Re} \sum_n \frac{\partial f(\phi)}{\partial \phi_n} \mathcal{F}_n \\
&\quad - 2\sqrt{2} \operatorname{Im} \sum_{Anm} (t_A)_{nm} (\bar{\psi}_{nL} \lambda_A) \phi_m + 2\sqrt{2} \operatorname{Im} \sum_{Anm} (t_A)_{mn} (\bar{\psi}_{nR} \lambda_A) \phi_m^* \\
&\quad - \sum_{Anm} \phi_n^* (t_A)_{nm} \phi_m D_A - \sum_A \xi_A D_A + \frac{1}{2} \sum_A D_A D_A \\
&\quad - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma}.
\end{aligned} \tag{27.4.1}$$

Here $f(\phi)$ is the superpotential, a gauge-invariant complex function of ϕ_n (but not of ϕ_n^*) which the condition of renormalizability requires to be a cubic polynomial; the ξ_A are constants which gauge invariance requires to vanish except where t_A is a $U(1)$ generator; the gauge-covariant derivatives are

$$D_\mu \psi_L \equiv \partial_\mu \psi_L - i \sum_A t_A V_{A\mu} \psi_L, \tag{27.4.2}$$

$$D_\mu \phi \equiv \partial_\mu \phi - i \sum_A t_A V_{A\mu} \phi, \tag{27.4.3}$$

$$(D_\mu \lambda)_A = \partial_\mu \lambda_A + \sum_{BC} C_{ABC} V_{B\mu} \lambda_C, \tag{27.4.4}$$

and $f_{A\mu\nu}$ is the gauge-covariant gauge field-strength tensor

$$f_{A\mu\nu} = \partial_\mu V_{A\nu} - \partial_\nu V_{A\mu} + \sum_{BC} C_{ABC} V_{B\mu} V_{C\nu}. \tag{27.4.5}$$

The auxiliary fields enter quadratically, with field-independent constants as the coefficients of the second-order term, so they can be eliminated by setting them equal to the values at which the Lagrangian density is stationary:

$$\mathcal{F}_n = - \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^*, \tag{27.4.6}$$

$$D_A = \xi_A + \sum_{nm} \phi_n^*(t_A)_{nm} \phi_m. \quad (27.4.7)$$

Using these back in Eq. (27.4.1), the Lagrangian density becomes

$$\begin{aligned} \mathcal{L} = & - \sum_n (D_\mu \phi)_n^* (D^\mu \phi)_n - \frac{1}{2} \sum_n (\bar{\psi}_{nL} \gamma^\mu (D_\mu \psi_L)_n) + \frac{1}{2} \sum_n \left(\overline{(D_\mu \psi_L)_n} \gamma^\mu \psi_{nL} \right) \\ & - \frac{1}{2} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} (\psi_{nL}^\top \epsilon \psi_{mL}) - \frac{1}{2} \sum_{nm} \left(\frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} \right)^* (\psi_{nL}^\top \epsilon \psi_{mL})^* \\ & - \sum_n \left| \frac{\partial f(\phi)}{\partial \phi_n} \right|^2 + i\sqrt{2} \sum_{Anm} (\bar{\psi}_{nL}(t_A)_{nm} \lambda_A) \phi_m \\ & - i\sqrt{2} \sum_{Anm} \phi_n^* (\bar{\lambda}_A(t_A)_{nm} \psi_{mL}) - \frac{1}{2} \sum_A \left(\xi_A + \sum_{nm} \phi_n^*(t_A)_{nm} \phi_m \right)^2 \\ & - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma}. \end{aligned} \quad (27.4.8)$$

Lorentz invariance requires the fields ψ_{nL} , λ_A , and $f_{A\mu\nu}$ to have vanishing vacuum expectation values, while the tree-value vacuum expectation values of the ϕ_n are at the minimum of the potential

$$V(\phi) = \sum_n \left| \frac{\partial f(\phi)}{\partial \phi_n} \right|^2 + \frac{1}{2} \sum_A \left(\xi_A + \sum_{nm} \phi_n^*(t_A)_{nm} \phi_m \right)^2. \quad (27.4.9)$$

This potential is positive, so if there is a set of field values at which $V(\phi)$ vanishes, then this is automatically also a minimum of the potential. For $V(\phi)$ to vanish at some field value $\phi_n = \phi_{n0}$, it is necessary and sufficient that

$$\mathcal{F}_{n0} = - \left[\frac{\partial f(\phi)}{\partial \phi_n} \right]_{\phi=\phi_0}^* = 0 \quad (27.4.10)$$

and

$$D_{A0} = \xi_A + \sum_{nm} \phi_{n0}^*(t_A)_{nm} \phi_{m0} = 0. \quad (27.4.11)$$

This in turn is the necessary and sufficient condition for supersymmetry not to be spontaneously broken, since Eq. (26.3.15) gives $\langle \delta \psi_{nL} \rangle_{\text{VAC}} = \sqrt{2} \langle \mathcal{F}_n \rangle_{\text{VAC}} \alpha_L$, and Eq. (26.2.16) gives $\langle \delta \lambda_A \rangle_{\text{VAC}} = i \langle D_A \rangle_{\text{VAC}} \gamma_5 \alpha$.

It is worth stressing here that spontaneous symmetry breaking is more difficult for supersymmetry than for other symmetries. For most symmetries of the action there will be field configurations at which the symmetry is unbroken and the potential is stationary, but the symmetry will nevertheless be spontaneously broken if none of these configurations are minima of the potential. In contrast, any supersymmetric field configuration gives the potential a value of zero, which is necessarily lower than the value of the potential for any non-supersymmetric configuration, so the existence of *any* supersymmetric field configuration insures that supersymmetry is unbroken. As we will see in Section 27.6, this

conclusion goes beyond the tree approximation used in this section; it is unaffected by corrections of any finite order in perturbation theory.

It may appear that Eqs. (27.4.10) and (27.4.11) impose too many conditions on the scalar fields to expect a solution, without some fine-tuning of the superpotential. However, for a gauge group of dimensionality D , the superpotential $f(\phi)$ is subject to the D constraints

$$\sum_m \frac{\partial f(\phi)}{\partial \phi_m} (t_A \phi)_m = 0, \quad (27.4.12)$$

for all A and all ϕ . Hence if ϕ has N independent components, then the number of *independent* conditions (27.4.10) is $N - D$, while the number of conditions (27.4.11) is D , so there are just N conditions altogether. With the number of conditions equal to the number of free variables, it is likely to find solutions for generic superpotentials. In fact, it is more usual to find solutions than not. For instance, for chiral scalar superfields in a non-trivial representation of a semi-simple gauge group we have $\xi_A = 0$, while $f(\phi)$ can have no terms linear in the ϕ_n , so both Eqs. (27.4.10) and (27.4.11) are satisfied for $\phi_{n0} = 0$. There may be other solutions of Eqs. (27.4.10) and (27.4.11) which break the gauge symmetries, but in such a theory supersymmetry cannot be broken, at least not in the tree approximation, and, as we shall see in Section 27.6, not in any order of perturbation theory.

More generally, it is easy to see that, even if the gauge group has $U(1)$ factors and even if the superpotential involves gauge-invariant superfields, if there exists a set of scalar field values ϕ_{n0} that satisfy Eq. (27.4.10), then there is another that satisfies both Eq. (27.4.10) and Eq. (27.4.11), provided only that the Fayet–Iliopoulos constants ξ_A all vanish. To show this, we note that since the superpotential $f(\phi)$ does not involve ϕ^* , it is invariant not only under ordinary gauge transformations $\phi \rightarrow \exp(i \sum_A \Lambda_A t_A) \phi$, with Λ_A arbitrary real numbers, but also under transformations with Λ_A arbitrary complex numbers. Under all these transformations the $\mathcal{F}$ -terms in Eq. (27.4.10) transform linearly, so if ϕ_0 satisfies Eq. (27.4.10), then so does $\phi^\Lambda \equiv \exp(i \sum_A \Lambda_A t_A) \phi_0$. On the other hand, the scalar product $[\phi^\dagger \phi]$ is not invariant under transformations with Λ_A complex, but $[\phi^{\Lambda\dagger} \phi^\Lambda]$ does remain real and positive for complex Λ_A , so it is bounded below, and therefore has a minimum. For $\xi_A = 0$, the condition that $[\phi^{\Lambda\dagger} \phi^\Lambda]$ be stationary at this minimum is just that ϕ^Λ should satisfy Eq. (27.4.11). We see then that in the absence of Fayet–Iliopoulos D -terms, the question of whether supersymmetry is unbroken in gauge theories is entirely a matter of whether the superpotential allows solutions of Eq. (27.4.10). The same result applies even in non-renormalizable theories [3].

Now let us assume that there is a set of values ϕ_{n0} at which $V(\phi_0) = 0$, so that supersymmetry is unbroken. The spin 0 degrees of freedom are described by a shifted field

$$\varphi_n = \phi_n - \phi_{n0}. \quad (27.4.13)$$

There is then a cross-term between φ and the gauge fields, arising from the first term in Eq. (27.4.1):

$$2 \sum_{nA} \text{Im} (\partial_\mu \varphi_n (t_A \phi_0)_n^*) V_A^\mu.$$

As shown in Section 21.1, it is always possible to eliminate this term by adopting a ‘unitarity gauge,’ in which ϕ_n satisfies a constraint that makes this term vanish:

$$\sum_n \text{Im}(\phi_n(t_A\phi_0)_n^*) = 0. \quad (27.4.14)$$

This will have the effect of eliminating the Goldstone bosons associated with broken gauge symmetries.

We shall now work out the masses of the particles of spin 0, spin 1/2, and spin 1 that arise in this theory if supersymmetry is unbroken, taking into account the possibility that the gauge symmetries may be spontaneously broken.

Spin 0

Because $\partial f(\phi)/\partial\phi_n$ and $\xi_A + \sum_{nm} \phi_n^*(t_A)_{nm}\phi_m$ must both vanish at $\phi_n = \phi_{n0}$, the terms in $V(\phi)$ of second order in $\varphi_n \equiv \phi_n - \phi_{n0}$ and/or φ_n^* are of the form

$$\begin{aligned} V_{\text{quad}}(\phi) = & \sum_{nm} (\mathcal{M}^* \mathcal{M})_{nm} \varphi_n^* \varphi_m + \sum_{Anm} (t_A\phi_0)_n (t_A\phi_0)_m^* \varphi_n^* \varphi_m \\ & + \frac{1}{2} \sum_{Anm} (t_A\phi_0)_n^* (t_A\phi_0)_m^* \varphi_n \varphi_m + \frac{1}{2} \sum_{Anm} (t_A\phi_0)_n (t_A\phi_0)_m \varphi_n^* \varphi_m^*, \end{aligned} \quad (27.4.15)$$

where $\mathcal{M}$ is the complex symmetric matrix (26.4.11):

$$\mathcal{M}_{nm} \equiv \left(\frac{\partial^2 f(\phi)}{\partial\phi_n \partial\phi_m} \right)_{\phi=\phi_0}.$$

This can be written as

$$V_{\text{quad}} = \frac{1}{2} \begin{bmatrix} \varphi \\ \varphi^* \end{bmatrix}^\dagger M_0^2 \begin{bmatrix} \varphi \\ \varphi^* \end{bmatrix}, \quad (27.4.16)$$

where M_0^2 is the block matrix

$$M_0^2 = \begin{bmatrix} \mathcal{M}^* \mathcal{M} + \sum_A (t_A\phi_0)(t_A\phi_0)^\dagger & \sum_A (t_A\phi_0)(t_A\phi_0)^T \\ \sum_A (t_A\phi_0)^*(t_A\phi_0)^\dagger & \mathcal{M} \mathcal{M}^* + \sum_A (t_A\phi_0)^*(t_A\phi_0)^T \end{bmatrix}. \quad (27.4.17)$$

Now we must look for the eigenvalues of this mass-squared matrix. Differentiating Eq. (27.4.12) with respect to ϕ_n gives

$$\sum_m \frac{\partial^2 f(\phi)}{\partial\phi_n \partial\phi_m} (t_A\phi)_m + \sum_m \frac{\partial f(\phi)}{\partial\phi_m} (t_A)_{mn} = 0. \quad (27.4.18)$$

But as we have seen, $\partial f(\phi)/\partial\phi_m$ vanishes at $\phi = \phi_0$, so by setting ϕ at this value in Eq. (27.4.18), we find

$$\sum_m \mathcal{M}_{nm} (t_A\phi_0)_m = 0. \quad (27.4.19)$$

It follows that

$$M_0^2 \begin{bmatrix} t_B\phi_0 \\ \pm(t_B\phi_0)^* \end{bmatrix} = \sum_A \left(\phi_0^\dagger [t_A t_B \pm t_B t_A] \phi_0 \right) \begin{bmatrix} t_B\phi_0 \\ \pm(t_B\phi_0)^* \end{bmatrix}.$$

But the vanishing of D_A at $\phi = \phi_0$ and the global gauge invariance of ξ_A tell us that

$$\left(\phi_0^\dagger[t_A, t_B]\phi_0\right) = i \sum_C C_{ABC} \left(\phi_0^\dagger t_C \phi_0\right) = -i \left(\phi_0^\dagger \phi_0\right) \sum_C C_{ABC} \xi_C = 0. \quad (27.4.20)$$

The matrix (27.4.17) therefore has a pair of eigenvectors for each gauge symmetry

$$u = \begin{bmatrix} \sum_B c_B t_B \phi_0 \\ \sum_B c_B (t_B \phi_0)^* \end{bmatrix}, \quad v = \begin{bmatrix} \sum_B c_B t_B \phi_0 \\ -\sum_B c_B (t_B \phi_0)^* \end{bmatrix}, \quad (27.4.21)$$

for which

$$M_0^2 u = \mu^2 u, \quad M_0^2 v = 0, \quad (27.4.22)$$

where μ^2 and c_A are any real solutions of the eigenvalue problem²

$$\sum_B \left(\phi_0^\dagger \{t_A, t_B\} \phi_0\right) c_B = \mu^2 c_A, \quad (27.4.23)$$

with the exception that if the eigenvalue μ^2 vanishes then $\sum_B c_B t_B \phi_0 = 0$, so that the eigenvectors u and v are absent. The massless particles associated with the v eigenvectors are Goldstone bosons, which are eliminated from the physical spectrum by the unitarity gauge condition (27.4.14). In addition to these mass eigenstates, there is another set that are orthogonal to all the u s and v s and that therefore take the form

$$w_\pm = \begin{bmatrix} \zeta \\ \pm \zeta^* \end{bmatrix}, \quad (27.4.24)$$

where

$$\sum_n (t_A \phi_0)_n^* \zeta_n = 0. \quad (27.4.25)$$

Eq. (27.4.19) shows that the space of ζ s satisfying Eq. (27.4.25) is invariant under multiplication with the Hermitian matrix $\mathcal{M}^\dagger \mathcal{M}$, so it is spanned by the eigenvectors of this matrix, satisfying

$$\mathcal{M}^\dagger \mathcal{M} \zeta = m^2 \zeta \quad (27.4.26)$$

with m^2 a set of real positive (or zero) eigenvalues. Eq. (27.4.26) and its complex conjugate together with Eq. (27.4.25) show that $w_\pm$ are eigenvectors of M_0^2 with eigenvalues m^2 :

$$M_0^2 w_\pm = m^2 w_\pm. \quad (27.4.27)$$

We thus have *two* self-charge-conjugate spinless bosons of each mass m satisfying Eq. (27.4.27), and one self-charge-conjugate spinless boson of each non-zero mass μ satisfying Eq. (27.4.23).

²The presence of a factor 1/2 in Eq. (21.1.17) which does not appear in Eq. (27.4.23) is due to a difference in the way that the scalar fields are normalized.

Spin 1/2

The fermion masses arise from the non-derivative terms in Eq. (27.4.8) that are of second order in the fermion fields ψ_n and λ_A :

$$\mathcal{L}_{1/2} = -\frac{1}{2} \sum_{nm} \mathcal{M}_{nm} (\psi_{nL}^T \epsilon \psi_{mL}) - i\sqrt{2} \sum_{Am} (t_A \phi_0)_m^* (\lambda_{LA}^T \epsilon \psi_{mL}) + \text{H.c.} \quad (27.4.28)$$

We saw in Section 26.4 that if the fermion mass term in the Lagrangian for a column χ of Majorana spinor fields is put in the form

$$\mathcal{L}_{1/2} = -\frac{1}{2} (\chi_L^T \epsilon M \chi_L) + \text{H.c.}, \quad (27.4.29)$$

then the fermion squared masses are the eigenvalues of the Hermitian matrix $M^\dagger M$. Eq. (27.4.28) gives the elements of the matrix M here as

$$M_{nm} = \mathcal{M}_{nm}, \quad M_{nA} = M_{An} = i\sqrt{2} (t_A \phi_0)_n^*, \quad M_{AB} = 0, \quad (27.4.30)$$

for which, using Eqs. (27.4.19) and (27.4.20),

$$\begin{aligned} (M^\dagger M)_{nm} &= (\mathcal{M}^\dagger \mathcal{M})_{nm} + 2 \sum_A (t_A \phi_0)_n (t_A \phi_0)_m^*, \\ (M^\dagger M)_{nA} &= (M^\dagger M)_{An} = 0, \\ (M^\dagger M)_{AB} &= 2 (\phi_0^\dagger t_B t_A \phi_0) = (\phi_0^\dagger \{t_B, t_A\} \phi_0). \end{aligned} \quad (27.4.31)$$

The eigenvectors of the matrix (27.4.30) are of three types. First are those of the form

$$z = \begin{bmatrix} \zeta \\ 0 \end{bmatrix}, \quad (27.4.32)$$

with eigenvalues m^2 , where ζ_n and m^2 are any eigenvectors and corresponding eigenvalues of $\mathcal{M}^\dagger \mathcal{M}$. Next are those of the form

$$g = \begin{bmatrix} 0 \\ c \end{bmatrix}, \quad (27.4.33)$$

with eigenvalues μ^2 , where c_B and μ^2 are any eigenvectors and eigenvalues of the matrix $(\phi_0^\dagger \{t_B, t_A\} \phi_0)$. Finally, there are those of the form

$$h = \begin{bmatrix} \sum_B c_B t_B \phi_0 \\ 0 \end{bmatrix}, \quad (27.4.34)$$

with eigenvalues μ^2 , where c_B and μ^2 are again any eigenvectors and eigenvalues of the matrix $(\phi_0^\dagger \{t_B, t_A\} \phi_0)$. The only exception is that the eigenvectors c of this matrix with eigenvalue zero have $\sum_A c_A t_A \phi_0 = 0$, corresponding to unbroken symmetries, so that in this case the vector (27.4.34) vanishes and we have only the eigenvector (27.4.33). Thus there is one Majorana fermion of each mass m satisfying Eq. (27.4.26), two Majorana fermions of each non-zero mass μ satisfying Eq. (27.4.22), and one Majorana fermion of zero mass for each unbroken gauge symmetry.

Spin 1

The mass terms in the Lagrangian for the gauge fields arise from the part of the first term in Eq. (27.4.1) that is of second order in the gauge field V_A^μ :

$$\mathcal{L}_V = - \sum_{nAB} (t_A \phi_0)_n^* (t_B \phi_0)_n V_{A\mu} V_B^\mu. \quad (27.4.35)$$

Since the fields $V_{A\mu}$ are real, their mass-squared matrix is the matrix in Eq. (27.4.23):

$$(\mu^2)_{AB} = \left(\phi_0^\dagger \{t_B, t_A\} \phi_0 \right). \quad (27.4.36)$$

There is one spin 1 particle of mass μ for each eigenvalue μ^2 of the matrix (27.4.36).

Putting this all together, we see that for each eigenvalue m^2 of the matrix $\mathcal{M}^* \mathcal{M}$ there are two self-charge-conjugate spinless particles and one Majorana fermion of mass m ; for each non-zero eigenvalue of the matrix μ_{AB}^2 there are one self-charge-conjugate spinless boson, two Majorana fermions, and one self-charge-conjugate spin 1 boson of mass μ ; and for each zero eigenvalue of this matrix there is one Majorana fermion and one self-charge-conjugate spin 1 boson of zero mass. It is not surprising that the particle multiplets for each zero or non-zero mass are just the same as we found by direct use of the supersymmetry algebra in Sections 25.4 and 25.5. What is a bit surprising is that the masses of the gauge and chiral particles are unaffected by each other. The masses m that are given by eigenvalues of $(\mathcal{M}^* \mathcal{M})_{nm}$ and the particles with these masses are just what they would be in a theory of chiral superfields without gauge superfields, and the masses μ that are given by eigenvalues of μ_{AB}^2 and the particles with these masses are just what they would be in a theory of gauge superfields with no chiral superfields.

For future use in Section 27.9 we will now apply the method described in Section 26.7 to construct the supersymmetry current for the supersymmetric gauge Lagrangian (27.4.1). In the gauge used earlier, an infinitesimal supersymmetry transformation changes V_A , λ_A , and D_A by the amounts (27.3.4)–(27.3.6). Adding the Noether supersymmetry current given by Eq. (26.7.2) for these fields to the Noether current for ϕ_n , ψ_n , and $\mathcal{F}_n$ already given in Eq. (26.7.8), with derivatives replaced by gauge-invariant derivatives, gives the total Noether supersymmetry current:

$$\begin{aligned} N^\mu = & \sum_A f_A^{\mu\nu} \gamma_\nu \lambda_A - \frac{1}{8} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu \lambda_A - \frac{1}{2} i \sum_A D_A \gamma_5 \gamma^\mu \lambda_A \\ & + \frac{1}{\sqrt{2}} \sum_n [2(D^\mu \phi)_n^* \psi_{nL} + 2(D^\mu \phi)_n \psi_{nR} + (\not{D}\phi)_n \gamma^\mu \psi_{nR} \\ & + (\not{D}\phi)_n^* \gamma^\mu \psi_{nL} - \mathcal{F}_n \gamma^\mu \psi_{nR} - \mathcal{F}_n^* \gamma^\mu \psi_{nL}]. \end{aligned} \quad (27.4.37)$$

This is not the supersymmetry current, because the Lagrangian density is not invariant under supersymmetry; instead, its change is the derivative

$$\delta \mathcal{L} = \partial_\mu (\bar{\alpha} K^\mu), \quad (27.4.38)$$

where³

$$\begin{aligned}
K^\mu = & \frac{1}{2}i \sum_A \epsilon^{\rho\sigma\mu\nu} f_{A\rho\sigma} \gamma_\nu \gamma_5 \lambda_A + \frac{1}{8} \sum_A [\gamma^\rho, \gamma^\sigma] \gamma^\mu \lambda_A f_{A\rho\sigma} + \frac{1}{2}i \sum_A D_A \gamma_5 \gamma^\mu \lambda_A \\
& - i \sum_{Anm} (t_A)_{nm} \gamma_5 \gamma^\mu \lambda_A \phi_n^* \phi_m \\
& + \frac{1}{\sqrt{2}} \sum_n \gamma^\mu \left[-(\not{D}\phi)_n \psi_{nR} - (\not{D}\phi)_n^* \psi_{nL} + \mathcal{F}_n^* \psi_{nL} + \mathcal{F}_n \psi_{nR} \right. \\
& \quad \left. + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^* \psi_{nL} + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right) \psi_{nR} \right].
\end{aligned} \tag{27.4.39}$$

The first two terms are derived using the identity (27.2.5). Using the same identity again with Eq. (26.7.4) gives the total supersymmetry current:

$$\begin{aligned}
S^\mu = N^\mu + K^\mu = & -\frac{1}{4} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu \lambda_A - i \sum_{Anm} (t_A)_{nm} \gamma_5 \gamma^\mu \lambda_A \phi_n^* \phi_m \\
& + \frac{1}{\sqrt{2}} \sum_n \left[(\not{D}\phi)_n \gamma^\mu \psi_{nR} + (\not{D}\phi)_n^* \gamma^\mu \psi_{nL} \right. \\
& \quad \left. + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right) \gamma^\mu \psi_{nL} + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^* \gamma^\mu \psi_{nR} \right].
\end{aligned} \tag{27.4.40}$$

* * *

In Section 26.8 we considered a class of supersymmetric theories with a superpotential $f(\Phi)$ that has an arbitrary dependence on a set of left-chiral scalar superfields Φ_n but not their derivatives, and with a Kahler potential $K(\Phi, \Phi^*)$ that has an arbitrary dependence on the Φ_n and Φ_n^* , but not their derivatives. We can extend the same considerations to gauge theories, again with the dependence of the Lagrangian on the chiral superfields limited only by supersymmetry, but without introducing new superderivatives or spacetime derivatives. The renormalizable Lagrangian density is then replaced with

$$\begin{aligned}
\mathcal{L} = & \frac{1}{2} \left[K \left(\Phi, \Phi^\dagger \exp \left(-2 \sum_A t_A V_A \right) \right) \right]_D + 2 \operatorname{Re}[f(\Phi)]_{\mathcal{F}} \\
& - \frac{1}{2} \operatorname{Re} \sum_{AB} [h_{AB}(\Phi) (W_{AL}^T \epsilon W_{BL})]_{\mathcal{F}},
\end{aligned} \tag{27.4.41}$$

where $h_{AB}(\Phi)$ is a new function of the Φ_n , but not of the Φ_n^* or derivatives.

The chiral gauge and scalar superfields are given by the expansions (26.3.21) and (27.3.15):

$$\begin{aligned}
W_{AL}(x, \theta) = & \lambda_{AL}(x_+) + \frac{1}{2} \gamma^\mu \gamma^\nu \theta_L f_{A\mu\nu}(x_+) + (\theta_L^T \epsilon \theta_L) \not{\partial} \lambda_{AR}(x_+) - i \theta_L D_A(x_+), \\
\Phi_n(x, \theta) = & \phi_n(x_+) - \sqrt{2} (\theta_L^T \epsilon \psi_{nL}(x_+)) + \mathcal{F}_n(x_+) (\theta_L^T \epsilon \theta_L),
\end{aligned}$$

³The easiest way to calculate the change in the term $[\Phi^\dagger \exp(-2 \sum_A t_A V_A) \Phi]_D$ is to calculate the λ -component of $\Phi^\dagger \exp(-2 \sum_A t_A V_A) \Phi$ and use Eq. (26.2.17). In this way of doing the calculation, the important term on the second line of the right-hand side of Eq. (27.4.39) arises from the λ -component of $\exp(-2 \sum_A t_A V_A)$.

where x_+^μ is the shifted coordinate (26.3.23). The terms of second order in θ_L (and independent of θ_R) in $\sum_{AB} h_{AB}(\Phi)(W_{AL}^T \epsilon W_{BL})$ are then

$$\begin{aligned}
-\left[\sum_{AB} h_{AB}(\Phi) (W_{AL}^T \epsilon W_{BL}) \right]_{\theta_L^2} &= (\theta_L^T \epsilon \theta_L) \sum_{AB} (\lambda_{AL}^T \epsilon \lambda_{BL}) \left[\frac{1}{2} \sum_{nm} (\psi_{nL}^T \epsilon \psi_{mL}) \frac{\partial^2 h_{AB}(\phi)}{\partial \phi_n \partial \phi_m} \right. \\
&\quad \left. - \sum_n \mathcal{F}_n \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \right] \\
&\quad + (\theta_L^T \epsilon \theta_L) \sum_{AB} h_{AB}(\phi) \left[-(\bar{\lambda}_A \not{\partial} (1 - \gamma_5) \lambda_B) - \frac{1}{2} f_{A\mu\nu} f_B^{\mu\nu} \right. \\
&\quad \left. + \frac{i}{4} \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_B^{\rho\sigma} + D_A D_B \right] \\
&\quad + \sqrt{2} \sum_{ABn} \frac{\partial h_{AB}(\phi)}{\partial \phi_n} (\theta_L^T \epsilon \psi_{nL}) \left[-(\lambda_{BL}^T \epsilon \gamma^\mu \gamma^\nu \theta_L) f_{A\mu\nu} \right. \\
&\quad \left. + 2i (\lambda_{BL}^T \epsilon \theta_L) \right],
\end{aligned}$$

with all fields now understood to be evaluated at x^μ rather than x_+^μ . (The first and second terms on the right-hand side are taken from Eqs. (26.4.4) and (27.3.16), respectively.) Also, by writing $\theta_{L\alpha} \theta_{L\beta}$ as $\frac{1}{2} \epsilon_{\alpha\beta} (\theta_L^T \epsilon \theta_L)$, the third term on the right-hand side can also be expressed as proportional to $(\theta_L^T \epsilon \theta_L)$:

$$\begin{aligned}
&(\theta_L^T \epsilon \psi_{nL}) [(\bar{\psi}_B \gamma^\mu \gamma^\nu \theta_L) f_{A\mu\nu} - 2i (\bar{\psi}_B \theta_L)] \\
&= \frac{1}{2} (\theta_L^T \epsilon \theta_L) [(\bar{\psi}_B \gamma^\mu \gamma^\nu \psi_{nL}) - 2i (\bar{\psi}_B \psi_{nL}) D_A].
\end{aligned}$$

The $\mathcal{F}$ -term is the coefficient of $(\theta_L^T \epsilon \theta_L)$, so

$$\begin{aligned}
-\left[\sum_{AB} h_{AB}(\Phi) (W_{AL}^T \epsilon W_{BL}) \right]_{\mathcal{F}} &= \sum_{AB} (\lambda_{AL}^T \epsilon \lambda_{BL}) \left[\frac{1}{2} \sum_{nm} (\psi_{nL}^T \epsilon \psi_{mL}) \frac{\partial^2 h_{AB}(\phi)}{\partial \phi_n \partial \phi_m} \right. \\
&\quad \left. - \sum_n \mathcal{F}_n \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \right] \\
&\quad + \sum_{AB} h_{AB}(\phi) \left[-(\bar{\lambda}_A \not{\partial} (1 - \gamma_5) \lambda_B) - \frac{1}{2} f_{A\mu\nu} f_B^{\mu\nu} \right. \\
&\quad \left. + \frac{i}{4} \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_B^{\rho\sigma} + D_A D_B \right] \\
&\quad + \frac{\sqrt{2}}{2} \sum_{ABn} \frac{\partial h_{AB}(\phi)}{\partial \phi_n} [-(\bar{\psi}_B \gamma^\mu \gamma^\nu \psi_{nL}) f_{A\mu\nu} + 2i (\bar{\psi}_B \psi_{nL}) D_A].
\end{aligned}$$

The other terms in Eq. (27.4.41) are just given by the gauge-invariant version of the La-

grangian density (26.8.6). Putting this together gives the Lagrangian density

$$\begin{aligned}
\mathcal{L} = & \operatorname{Re} \sum_{nm} \mathcal{G}_{nm}(\phi, \phi^*) \left[-\frac{1}{2} (\bar{\psi}_m \not{D} (1 + \gamma_5) \psi_n) + \mathcal{F}_n \mathcal{F}_m^* - D_\mu \phi_n D^\mu \phi_m^* \right] \\
& - 2 \operatorname{Re} \sum_i \frac{\partial K(\phi, \phi^*)}{\partial \phi_i^*} D_A (\phi^* t_A)_i \\
& + i\sqrt{2} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_i \partial \phi_j^*} [(t_A \phi)_i \bar{\psi}_j \lambda_{AR} - (\phi^* t_A)_j \bar{\psi}_i \lambda_{AL}] \\
& - \operatorname{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_n \psi_{mL}) \mathcal{F}_\ell^* \\
& + \operatorname{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_m \gamma^\mu \psi_{\ell R}) D_\mu \phi_n \\
& + \frac{1}{4} \sum_{nm\ell k} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^* \partial \phi_k^*} (\bar{\psi}_n \psi_{mL}) (\bar{\psi}_k \psi_{\ell R}) \\
& - \operatorname{Re} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} (\bar{\psi}_n \psi_{mL}) + 2 \operatorname{Re} \sum_n \mathcal{F}_n \frac{\partial f(\phi)}{\partial \phi_n} \\
& + \frac{1}{4} \operatorname{Re} \sum_{ABnm} (\bar{\lambda}_A \lambda_{BL}) (\bar{\psi}_n \psi_{mL}) \frac{\partial^2 h_{AB}(\phi)}{\partial \phi_n \partial \phi_m} \\
& - \frac{1}{2} \operatorname{Re} \sum_{ABn} (\bar{\lambda}_A \lambda_{BL}) \mathcal{F}_n \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \\
& + \operatorname{Re} \sum_{AB} h_{AB}(\phi) \left[-(\bar{\lambda}_A \not{D} \lambda_{BR}) - \frac{1}{4} f_{A\mu\nu} f_B^{\mu\nu} + \frac{i}{8} \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_B^{\rho\sigma} + \frac{1}{2} D_A D_B \right] \\
& + \frac{\sqrt{2}}{4} \operatorname{Re} \sum_{ABn} \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \left[-(\bar{\lambda}_B \gamma^\mu \gamma^\nu \psi_{nL}) f_{A\mu\nu} + 2i (\bar{\lambda}_B \psi_{nL}) D_A \right].
\end{aligned} \tag{27.4.42}$$

One interesting feature of this result is the appearance of a gaugino mass in theories with ϕ_n -dependent functions $h_{AB}(\phi)$, when supersymmetry is broken by a non-vanishing value of $\mathcal{F}_n$. This mechanism is used to generate gaugino masses in some theories of gravitationally mediated supersymmetry breaking, discussed in Section 31.7.